

## Regression Analysis

1. If the regression coefficients are 0.8 and 0.2, what would be the value of coefficient of correlation.

Ans: 0.4

2. Calculate linear regression coefficients from the following:

|   |   |   |    |    |    |    |    |    |
|---|---|---|----|----|----|----|----|----|
| x | 1 | 2 | 3  | 4  | 5  | 6  | 7  | 8  |
| y | 3 | 7 | 10 | 12 | 14 | 17 | 20 | 24 |

Ans:  $b_{yx} = 2.7976$        $b_{xy} = 0.3540$

3. The following table gives age (x) in years of cars and annual maintenance cost (y) in hundred rupees

|   |    |    |    |    |    |
|---|----|----|----|----|----|
| x | 1  | 3  | 5  | 7  | 9  |
| y | 15 | 18 | 21 | 23 | 22 |

Estimate the maintenance cost for a 4 years old car after finding the regression Eq<sup>n</sup>.

Ans: 1885 Rupees

4. In a partially destroyed laboratory record of an analysis of a correlation data, the following results only are legible:

Variance of  $x = 9$

Regression Equations:  $8x - 10y + 66 = 0$ ,  $40x - 18y = 214$ .

What were (I)  $\bar{x}$  and  $\bar{y}$  (II) the S.D. of  $y$

(III) Coefficient of correlation

Ans: (I)  $\bar{x} = 13$ ,  $\bar{y} = 17$

(II)  $\sigma_y = 4$

(III)  $r = 0.6$

5. The following results were obtained from records of age ( $x$ ) and blood pressure ( $y$ ) of a group of 10 men:

|          | $x$ | $y$ |
|----------|-----|-----|
| Mean     | 53  | 142 |
| Variance | 130 | 165 |

and  $\sum (x - \bar{x})(y - \bar{y}) = 1220$ . Find the appropriate regression Equation and use it to estimate the blood pressure of a man whose age is 45.

Ans:  $\hat{y} = 0.93846x + 92.26162$ , Required blood pressure = 134.49



6. The following results were obtained from marks in Applied mechanics and Engineering mathematics in an Examination:

|                    | Applied mechanics<br>( $x$ ) | Engg. Math<br>( $y$ ) |
|--------------------|------------------------------|-----------------------|
| Mean               | 47.5                         | 39.5                  |
| Standard Deviation | 16.8                         | 10.8                  |

$r = 0.95$ . Find both the regression Equations.

Also Estimate the value of  $y$  for  $x = 30$ .

Ans: Regression line of  $y$  on  $x$ ,  $y = 0.6107x + 10.49$

Regression line of  $x$  on  $y$   $x = 1.477y - 10.8415$

$y$ , when  $x = 30$  is 28.81

7. Two lines of regression are given by  $x + 2y - 5 = 0$  and  $2x + 3y - 8 = 0$  and  $\sigma_x^2 = 12$

Calculate (I)  $\bar{x}$  and  $\bar{y}$  (II) the coefficient of correlation  $r$  b/w  $x$  &  $y$  (III)  $\sigma_y$

Ans: (I)  $\bar{x} = 1$ ,  $\bar{y} = 2$  (II)  $r = -\frac{\sqrt{3}}{2}$  (III) 4



8. An analyst for a company was studying travelling expenses ( $y$ ) in rupees and duration ( $x$ ) of these trips for 102 Sales trip. He has found relation b/w  $x$  and  $y$  is linear and data as follows:

$$\Sigma x = 510 \quad \Sigma y = 7140 \quad \Sigma x^2 = 4150 \quad \Sigma xy = 54900$$

$$\Sigma y^2 = 740200$$

Calculate: (I) Two regression lines

(II) A given trip has to take 7 days. How much money should be allowed so that they will not run short of money?

Ans: (I)  $y = 12x + 10$      $x = 0.07986y - 0.59068$

(II) 94 Rupees.

9. If the coefficient of correlation b/w two variables  $x$  and  $y$  is 0.5 and acute angle b/w their line of regression is  $\tan^{-1}(\frac{3}{5})$ , then show that either

$$2\sigma_x = \sigma_y \text{ or } 2\sigma_y = \sigma_x$$

10. Can  $y = 5 + 2.8x$  and  $x = 3 - 0.5y$  be the estimated regression Equations of  $y$  on  $x$  and  $x$  on  $y$  respectively? Explain your answer with suitable theoretical arguments.

Ans: No,  $b_{yx} = 2.8$  and  $b_{xy} = -0.5$  which is not possible.

11. Given  $N=50$ , Mean of  $y = 44$ . Variance of  $x$  is  $\frac{9}{16}$  of the variance of  $y$ . Regression Equation of  $x$  on  $y$  is  $3y - 5x = -180$ . Find

(I) Mean of  $x$  (II) coefficient of correlation

between  $x$  and  $y$  (Note:  $\sigma_x^2 = \frac{9}{16} \sigma_y^2$  (A.T.Q.))

Ans: (I) 62.4 (II) 0.8

THE END